

Supplementary Material

A Family of Classifying Criteria Emerging from Least-Squares Matching Between Partition Incidence Matrices

Boris Mirkin, Azizbek Bekniyazov

Results of Computational Experiments

Synthetic data

Synthetic datasets used in the experiments.

Dataset	V	L	α
1	6	4	0.5
2	6	4	0.85
3	15	4	0.5
4	15	4	0.85
5	6	8	0.5
6	6	8	0.85
7	15	8	0.5
8	15	8	0.85
9	6	15	0.5
10	6	15	0.85
11	15	15	0.5
12	15	15	0.85

Table 1.1. $N = 2000$, $V = 6$, $L = 4$, $\alpha = 0.5$, tree_depth = 3

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l^{0.5} \log(p_l)$	$b = -p_l \log(p_l)$
Accuracy	Mean	0.9949	0.9937	0.9732	0.9912	0.8742	0.8526	0.9924	0.9935	0.9490
	Std	0.0158	0.0184	0.0733	0.0362	0.1505	0.1339	0.0190	0.0183	0.0971
Precision	Mean	0.9949	0.9938	0.9688	0.9920	0.8748	0.8463	0.9926	0.9936	0.9274
	Std	0.0158	0.0181	0.0926	0.0305	0.1696	0.1653	0.0187	0.0181	0.1421
Recall	Mean	0.9949	0.9937	0.9732	0.9912	0.8742	0.8526	0.9924	0.9935	0.9490
	Std	0.0158	0.0184	0.0733	0.0362	0.1505	0.1339	0.0190	0.0183	0.0971
F1 score	Mean	0.9949	0.9937	0.9654	0.9898	0.8358	0.8067	0.9925	0.9935	0.9327
	Std	0.0159	0.0183	0.0967	0.0457	0.1946	0.1745	0.0189	0.0182	0.1297
ARI	Mean	0.9875	0.9845	0.9643	0.9836	0.8440	0.8169	0.9813	0.9841	0.9360
	Std	0.0351	0.0412	0.0890	0.0494	0.1880	0.1676	0.0429	0.0410	0.1152

Table 1.2. $N = 2000$, $V = 6$, $L = 4$, $\alpha = 0.5$, tree_depth = 4

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l^{0.5} \log(p_l)$	$b = -p_l \log(p_l)$
Accuracy	Mean	0.9955	0.9952	0.9878	0.9921	0.9636	0.9530	0.9949	0.9951	0.9787
	Std	0.0146	0.0150	0.0480	0.0355	0.0850	0.0946	0.0151	0.0150	0.0646
Precision	Mean	0.9956	0.9952	0.9907	0.9929	0.9667	0.9496	0.9949	0.9951	0.9764
	Std	0.0145	0.0148	0.0346	0.0302	0.0853	0.1125	0.0148	0.0148	0.0775
Recall	Mean	0.9955	0.9952	0.9878	0.9921	0.9636	0.9530	0.9949	0.9951	0.9787
	Std	0.0146	0.0150	0.0480	0.0355	0.0850	0.0946	0.0151	0.0150	0.0646
F1 score	Mean	0.9955	0.9952	0.9849	0.9907	0.9535	0.9398	0.9949	0.9951	0.9725
	Std	0.0146	0.0149	0.0619	0.0452	0.1101	0.1232	0.0150	0.0149	0.0853
ARI	Mean	0.9890	0.9879	0.9812	0.9860	0.9521	0.9369	0.9872	0.9877	0.9711
	Std	0.0326	0.0350	0.0613	0.0467	0.1053	0.1198	0.0351	0.0349	0.0781

Table 2.1. $N = 2000$, $V = 6$, $L = 4$, $\alpha = 0.85$, tree_depth = 3

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l^{0.5} \log(p_l)$	$b = -p_l \log(p_l)$
Accuracy	Mean	0.7804	0.7828	0.7714	0.7806	0.5241	0.5155	0.7853	0.7842	0.7740
	Std	0.0686	0.0646	0.0752	0.0692	0.0471	0.0445	0.0648	0.0643	0.0683
Precision	Mean	0.7912	0.7938	0.7751	0.7855	0.4978	0.4750	0.7961	0.7953	0.7814
	Std	0.0654	0.0623	0.0841	0.0783	0.1042	0.1029	0.0611	0.0621	0.0788
Recall	Mean	0.7804	0.7828	0.7714	0.7806	0.5241	0.5155	0.7853	0.7842	0.7740
	Std	0.0686	0.0646	0.0752	0.0692	0.0471	0.0445	0.0648	0.0643	0.0683
F1 score	Mean	0.7785	0.7829	0.7622	0.7768	0.4309	0.4277	0.7858	0.7841	0.7671
	Std	0.0720	0.0651	0.0884	0.0769	0.0681	0.0652	0.0645	0.0654	0.0789
ARI	Mean	0.5261	0.5294	0.5301	0.5351	0.3093	0.2918	0.5316	0.5300	0.5269
	Std	0.1173	0.1111	0.1156	0.1094	0.0687	0.0695	0.1131	0.1123	0.1081

Table 2.2. $N = 2000$, $V = 6$, $L = 4$, $\alpha = 0.85$, tree_depth = 4

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l^{0.5} \log(p_l)$	$b = -p_l \log(p_l)$
Accuracy	Mean	0.8058	0.8060	0.8003	0.8066	0.5471	0.5536	0.8046	0.8064	0.7963
	Std	0.0614	0.0594	0.0636	0.0608	0.0584	0.0586	0.0619	0.0605	0.0631
Precision	Mean	0.8114	0.8111	0.8028	0.8111	0.5804	0.5657	0.8111	0.8125	0.8011
	Std	0.0590	0.0589	0.0672	0.0603	0.0872	0.0892	0.0588	0.0583	0.0659
Recall	Mean	0.8058	0.8060	0.8003	0.8066	0.5471	0.5536	0.8046	0.8064	0.7963
	Std	0.0614	0.0594	0.0636	0.0608	0.0584	0.0586	0.0619	0.0605	0.0631
F1 score	Mean	0.8047	0.8054	0.7958	0.8045	0.4733	0.4919	0.8045	0.8062	0.7923
	Std	0.0623	0.0600	0.0722	0.0639	0.0824	0.0761	0.0616	0.0603	0.0693
ARI	Mean	0.5734	0.5704	0.5687	0.5758	0.3217	0.3157	0.5661	0.5700	0.5630
	Std	0.1089	0.1068	0.1061	0.1064	0.0703	0.0832	0.1126	0.1097	0.1050

Table 3.1. $N = 2000$, $V = 15$, $L = 4$, $\alpha = 0.5$, tree_depth = 3

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l^{0.5} \log(p_l)$	$b = -p_l \log(p_l)$
Accuracy	Mean	0.9986	0.9980	0.9944	0.9981	0.9433	0.9124	0.9972	0.9976	0.9565
	Std	0.0022	0.0029	0.0327	0.0028	0.1163	0.1204	0.0040	0.0032	0.0901
Precision	Mean	0.9986	0.9980	0.9919	0.9981	0.9302	0.8837	0.9972	0.9977	0.9397
	Std	0.0022	0.0029	0.0499	0.0028	0.1424	0.1705	0.0039	0.0032	0.1299
Recall	Mean	0.9986	0.9980	0.9944	0.9981	0.9433	0.9124	0.9972	0.9976	0.9565
	Std	0.0022	0.0029	0.0327	0.0028	0.1163	0.1204	0.0040	0.0032	0.0901
F1 score	Mean	0.9986	0.9980	0.9927	0.9981	0.9257	0.8837	0.9972	0.9976	0.9424
	Std	0.0022	0.0029	0.0439	0.0028	0.1515	0.1605	0.0040	0.0032	0.1203
ARI	Mean	0.9963	0.9946	0.9918	0.9950	0.9284	0.8928	0.9925	0.9937	0.9475
	Std	0.0058	0.0076	0.0390	0.0074	0.1467	0.1453	0.0106	0.0085	0.1048

Table 3.2. $N = 2000$, $V = 15$, $L = 4$, $\alpha = 0.5$, tree_depth = 4

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l^{0.5} \log(p_l)$	$b = -p_l \log(p_l)$
Accuracy	Mean	0.9986	0.9987	0.9990	0.9984	0.9886	0.9883	0.9983	0.9984	0.9940
	Std	0.0022	0.0018	0.0015	0.0026	0.0496	0.0493	0.0026	0.0026	0.0312
Precision	Mean	0.9986	0.9987	0.9990	0.9984	0.9885	0.9881	0.9983	0.9984	0.9917
	Std	0.0022	0.0018	0.0014	0.0026	0.0561	0.0562	0.0025	0.0026	0.0473
Recall	Mean	0.9986	0.9987	0.9990	0.9984	0.9886	0.9883	0.9983	0.9984	0.9940
	Std	0.0022	0.0018	0.0015	0.0026	0.0496	0.0493	0.0026	0.0026	0.0312
F1 score	Mean	0.9986	0.9987	0.9990	0.9984	0.9853	0.9851	0.9983	0.9984	0.9924
	Std	0.0022	0.0018	0.0015	0.0026	0.0659	0.0650	0.0026	0.0026	0.0418
ARI	Mean	0.9963	0.9965	0.9974	0.9958	0.9843	0.9832	0.9956	0.9957	0.9906
	Std	0.0058	0.0048	0.0039	0.0068	0.0602	0.0614	0.0067	0.0068	0.0365

Table 4.1. $N = 2000$, $V = 15$, $L = 4$, $\alpha = 0.85$, tree_depth = 3

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l^{0.5} \log(p_l)$	$b = -p_l \log(p_l)$
Accuracy	Mean	0.8748	0.8774	0.8581	0.8752	0.5033	0.5055	0.8740	0.8757	0.8544
	Std	0.0384	0.0376	0.0594	0.0398	0.0189	0.0257	0.0380	0.0368	0.0703
Precision	Mean	0.8806	0.8839	0.8615	0.8813	0.4920	0.5028	0.8807	0.8820	0.8622
	Std	0.0363	0.0340	0.0658	0.0366	0.1418	0.1278	0.0353	0.0341	0.0727
Recall	Mean	0.8748	0.8774	0.8581	0.8752	0.5033	0.5055	0.8740	0.8757	0.8544
	Std	0.0384	0.0376	0.0594	0.0398	0.0189	0.0257	0.0380	0.0368	0.0703
F1 score	Mean	0.8752	0.8783	0.8529	0.8759	0.3876	0.3973	0.8750	0.8766	0.8505
	Std	0.0383	0.0371	0.0738	0.0395	0.0338	0.0441	0.0376	0.0364	0.0873
ARI	Mean	0.7005	0.7045	0.6839	0.7009	0.3114	0.3065	0.6969	0.7004	0.6726
	Std	0.0825	0.0813	0.0932	0.0854	0.0559	0.0575	0.0817	0.0793	0.1011

Table 4.2. $N = 2000$, $V = 15$, $L = 4$, $\alpha = 0.85$, tree_depth = 4

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l^{0.5} \log(p_l)$	$b = -p_l \log(p_l)$
Accuracy	Mean	0.9034	0.9035	0.8901	0.9020	0.5220	0.5612	0.8997	0.9007	0.8848
	Std	0.0336	0.0293	0.0473	0.0316	0.0414	0.0846	0.0330	0.0322	0.0655
Precision	Mean	0.9059	0.9060	0.8929	0.9047	0.6059	0.6101	0.9033	0.9045	0.8923
	Std	0.0322	0.0281	0.0459	0.0308	0.0983	0.1059	0.0311	0.0301	0.0494
Recall	Mean	0.9034	0.9035	0.8901	0.9020	0.5220	0.5612	0.8997	0.9007	0.8848
	Std	0.0336	0.0293	0.0473	0.0316	0.0414	0.0846	0.0330	0.0322	0.0655
F1 score	Mean	0.9036	0.9036	0.8873	0.9019	0.4176	0.4810	0.9000	0.9011	0.8813
	Std	0.0336	0.0294	0.0566	0.0318	0.0599	0.1079	0.0329	0.0321	0.0822
ARI	Mean	0.7636	0.7624	0.7437	0.7597	0.3272	0.3678	0.7532	0.7557	0.7337
	Std	0.0748	0.0655	0.0825	0.0707	0.0694	0.1146	0.0742	0.0724	0.0914

Table 5.1. $N = 2000$, $V = 6$, $L = 8$, $\alpha = 0.5$, tree_depth = 3

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l^{0.5} \log(p_l)$	$b = -p_l \log(p_l)$
Accuracy	Mean	0.9056	0.7183	0.6600	0.7268	0.5128	0.4734	0.7038	0.7200	0.5694
	Std	0.0578	0.1192	0.1488	0.1124	0.1656	0.1194	0.1219	0.1153	0.1502
Precision	Mean	0.8742	0.6361	0.5567	0.6328	0.4354	0.4037	0.6366	0.6440	0.4842
	Std	0.0881	0.1553	0.1757	0.1487	0.1690	0.1315	0.1548	0.1502	0.1704
Recall	Mean	0.9056	0.7183	0.6600	0.7268	0.5128	0.4734	0.7038	0.7200	0.5694
	Std	0.0578	0.1192	0.1488	0.1124	0.1656	0.1194	0.1219	0.1153	0.1502
F1 score	Mean	0.8802	0.6436	0.5711	0.6525	0.4096	0.3631	0.6278	0.6459	0.4699
	Std	0.0777	0.1440	0.1777	0.1390	0.1854	0.1341	0.1470	0.1395	0.1750
ARI	Mean	0.8765	0.6656	0.6070	0.6740	0.4197	0.3770	0.6451	0.6676	0.4895
	Std	0.0645	0.1581	0.1954	0.1434	0.2135	0.1618	0.1649	0.1536	0.1924

Table 5.2. $N = 2000$, $V = 6$, $L = 8$, $\alpha = 0.5$, tree_depth = 4

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l^{0.5} \log(p_l)$	$b = -p_l \log(p_l)$
Accuracy	Mean	0.9813	0.8988	0.8387	0.9020	0.6319	0.5989	0.8884	0.9005	0.7221
	Std	0.0191	0.1020	0.1451	0.0878	0.1920	0.1607	0.1078	0.0996	0.1649
Precision	Mean	0.9806	0.8667	0.8088	0.8750	0.6010	0.5809	0.8710	0.8705	0.6894
	Std	0.0257	0.1375	0.1655	0.1153	0.2040	0.1681	0.1370	0.1315	0.1776
Recall	Mean	0.9813	0.8988	0.8387	0.9020	0.6319	0.5989	0.8884	0.9005	0.7221
	Std	0.0191	0.1020	0.1451	0.0878	0.1920	0.1607	0.1078	0.0996	0.1649
F1 score	Mean	0.9806	0.8710	0.7978	0.8754	0.5490	0.5151	0.8598	0.8734	0.6542
	Std	0.0234	0.1333	0.1818	0.1146	0.2276	0.1846	0.1393	0.1302	0.2008
ARI	Mean	0.9615	0.8767	0.7981	0.8763	0.5520	0.4992	0.8593	0.8791	0.6596
	Std	0.0301	0.1101	0.1787	0.1017	0.2363	0.2111	0.1204	0.1054	0.2056

Table 6.1. $N = 2000$, $V = 6$, $L = 8$, $\alpha = 0.85$, tree_depth = 3

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l^{0.5} \log(p_l)$	$b = -p_l \log(p_l)$
Accuracy	Mean	0.5639	0.5518	0.5150	0.5396	0.2947	0.2945	0.5498	0.5520	0.4697
	Std	0.0552	0.0667	0.0671	0.0687	0.0314	0.0343	0.0650	0.0661	0.0769
Precision	Mean	0.5172	0.5211	0.4289	0.4854	0.2233	0.2253	0.5217	0.5241	0.3800
	Std	0.0806	0.0901	0.0862	0.0867	0.0736	0.0766	0.0801	0.0808	0.1095
Recall	Mean	0.5639	0.5518	0.5150	0.5396	0.2947	0.2945	0.5498	0.5520	0.4697
	Std	0.0552	0.0667	0.0671	0.0687	0.0314	0.0343	0.0650	0.0661	0.0769
F1 score	Mean	0.5210	0.5091	0.4400	0.4813	0.1817	0.1929	0.5080	0.5106	0.3793
	Std	0.0676	0.0794	0.0801	0.0840	0.0352	0.0389	0.0748	0.0755	0.0934
ARI	Mean	0.3433	0.3244	0.3236	0.3316	0.1565	0.1465	0.3162	0.3223	0.2865
	Std	0.0640	0.0710	0.0687	0.0738	0.0370	0.0381	0.0728	0.0715	0.0754

Table 6.2. $N = 2000$, $V = 6$, $L = 8$, $\alpha = 0.85$, tree_depth = 4

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l^{0.5} \log(p_l)$	$b = -p_l \log(p_l)$
Accuracy	Mean	0.6254	0.6288	0.5907	0.6160	0.3206	0.3255	0.6177	0.6248	0.5402
	Std	0.0607	0.0627	0.0630	0.0671	0.0446	0.0365	0.0657	0.0609	0.0767
Precision	Mean	0.6421	0.6494	0.5751	0.6127	0.3001	0.3141	0.6408	0.6476	0.5140
	Std	0.0621	0.0654	0.0811	0.0836	0.0894	0.0961	0.0638	0.0612	0.1012
Recall	Mean	0.6254	0.6288	0.5907	0.6160	0.3206	0.3255	0.6177	0.6248	0.5402
	Std	0.0607	0.0627	0.0630	0.0671	0.0446	0.0365	0.0657	0.0609	0.0767
F1 score	Mean	0.6178	0.6229	0.5549	0.5954	0.2218	0.2410	0.6132	0.6210	0.4844
	Std	0.0641	0.0689	0.0745	0.0793	0.0511	0.0428	0.0697	0.0649	0.0945
ARI	Mean	0.3809	0.3744	0.3727	0.3807	0.1703	0.1612	0.3589	0.3675	0.3354
	Std	0.0710	0.0750	0.0703	0.0742	0.0390	0.0365	0.0818	0.0737	0.0771

Table 7.1. $N = 2000$, $V = 15$, $L = 8$, $\alpha = 0.5$, tree_depth = 3

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l^{0.5} \log(p_l)$	$b = -p_l \log(p_l)$
Accuracy	Mean	0.9568	0.6850	0.6879	0.7172	0.4886	0.4218	0.6469	0.6776	0.6642
	Std	0.0516	0.1053	0.1103	0.0939	0.1399	0.0929	0.1072	0.1043	0.1128
Precision	Mean	0.9388	0.5883	0.5727	0.6104	0.3922	0.3573	0.5717	0.5822	0.5454
	Std	0.0793	0.1265	0.1464	0.1232	0.1456	0.1079	0.1273	0.1221	0.1423
Recall	Mean	0.9568	0.6850	0.6879	0.7172	0.4886	0.4218	0.6469	0.6776	0.6642
	Std	0.0516	0.1053	0.1103	0.0939	0.1399	0.0929	0.1072	0.1043	0.1128
F1 score	Mean	0.9448	0.6052	0.6028	0.6396	0.3853	0.3219	0.5688	0.5969	0.5757
	Std	0.0697	0.1247	0.1357	0.1145	0.1537	0.0951	0.1246	0.1230	0.1344
ARI	Mean	0.9453	0.6202	0.6330	0.6654	0.3680	0.2603	0.5420	0.6097	0.6081
	Std	0.0531	0.1431	0.1433	0.1212	0.1783	0.1251	0.1493	0.1413	0.1450

Table 7.2. $N = 2000$, $V = 15$, $L = 8$, $\alpha = 0.5$, tree_depth = 4

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l^{0.5} \log(p_l)$	$b = -p_l \log(p_l)$
Accuracy	Mean	0.9952	0.8648	0.8790	0.8971	0.6499	0.5176	0.8052	0.8575	0.8512
	Std	0.0046	0.0966	0.0984	0.0854	0.1659	0.1415	0.0969	0.0969	0.1288
Precision	Mean	0.9954	0.8208	0.8262	0.8479	0.5768	0.4959	0.7712	0.8172	0.7962
	Std	0.0044	0.1329	0.1413	0.1248	0.1882	0.1404	0.1238	0.1313	0.1716
Recall	Mean	0.9952	0.8648	0.8790	0.8971	0.6499	0.5176	0.8052	0.8575	0.8512
	Std	0.0046	0.0966	0.0984	0.0854	0.1659	0.1415	0.0969	0.0969	0.1288
F1 score	Mean	0.9952	0.8265	0.8417	0.8642	0.5693	0.4304	0.7567	0.8178	0.8104
	Std	0.0046	0.1220	0.1275	0.1121	0.1948	0.1522	0.1173	0.1225	0.1597
ARI	Mean	0.9891	0.8403	0.8594	0.8893	0.5597	0.3665	0.7545	0.8309	0.8207
	Std	0.0104	0.1189	0.1166	0.0916	0.2134	0.1908	0.1344	0.1194	0.1651

Table 8.1. $N = 2000$, $V = 15$, $L = 8$, $\alpha = 0.85$, tree_depth = 3

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l^{0.5} \log(p_l)$	$b = -p_l \log(p_l)$
Accuracy	Mean	0.6706	0.6427	0.5803	0.6296	0.2702	0.2844	0.6356	0.6406	0.4637
	Std	0.0392	0.0583	0.0751	0.0619	0.0267	0.0240	0.0587	0.0565	0.0839
Precision	Mean	0.6412	0.6095	0.4898	0.5949	0.2123	0.2340	0.6038	0.6049	0.3624
	Std	0.0719	0.0803	0.1124	0.0958	0.0942	0.0767	0.0732	0.0770	0.1047
Recall	Mean	0.6706	0.6427	0.5803	0.6296	0.2702	0.2844	0.6356	0.6406	0.4637
	Std	0.0392	0.0583	0.0751	0.0619	0.0267	0.0240	0.0587	0.0565	0.0839
F1 score	Mean	0.6381	0.6016	0.5023	0.5803	0.1470	0.1783	0.5952	0.5993	0.3582
	Std	0.0543	0.0728	0.0983	0.0805	0.0325	0.0323	0.0732	0.0717	0.1056
ARI	Mean	0.4695	0.4411	0.4109	0.4333	0.1479	0.1482	0.4323	0.4388	0.3089
	Std	0.0428	0.0619	0.0698	0.0665	0.0298	0.0353	0.0617	0.0614	0.0769

Table 8.2. $N = 2000$, $V = 15$, $L = 8$, $\alpha = 0.85$, tree_depth = 4

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l^{0.5} \log(p_l)$	$b = -p_l \log(p_l)$
Accuracy	Mean	0.7804	0.7606	0.6891	0.7479	0.2850	0.3080	0.7503	0.7556	0.5624
	Std	0.0346	0.0496	0.0662	0.0533	0.0389	0.0319	0.0462	0.0481	0.1007
Precision	Mean	0.7979	0.7782	0.6715	0.7652	0.2830	0.3075	0.7706	0.7716	0.5456
	Std	0.0305	0.0496	0.0896	0.0524	0.0965	0.0820	0.0438	0.0515	0.1188
Recall	Mean	0.7804	0.7606	0.6891	0.7479	0.2850	0.3080	0.7503	0.7556	0.5624
	Std	0.0346	0.0496	0.0662	0.0533	0.0389	0.0319	0.0462	0.0481	0.1007
F1 score	Mean	0.7807	0.7568	0.6546	0.7374	0.1681	0.2147	0.7476	0.7514	0.4950
	Std	0.0346	0.0534	0.0867	0.0624	0.0446	0.0382	0.0496	0.0532	0.1257
ARI	Mean	0.5760	0.5541	0.5036	0.5471	0.1616	0.1619	0.5375	0.5468	0.3855
	Std	0.0577	0.0697	0.0674	0.0684	0.0365	0.0367	0.0633	0.0680	0.0984

Table 9.1. $N = 2000$, $V = 6$, $L = 15$, $\alpha = 0.5$, tree_depth = 3

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l^{0.5} \log(p_l)$	$b = -p_l \log(p_l)$
Accuracy	Mean	0.5365	0.4661	0.4130	0.4752	0.2081	0.2372	0.4513	0.4665	0.2844
	Std	0.0116	0.0641	0.0776	0.0624	0.0575	0.0470	0.0656	0.0651	0.0868
Precision	Mean	0.3339	0.3349	0.2770	0.3321	0.1510	0.1743	0.3370	0.3337	0.1976
	Std	0.0153	0.0605	0.0694	0.0553	0.0605	0.0486	0.0585	0.0655	0.0775
Recall	Mean	0.5365	0.4661	0.4130	0.4752	0.2081	0.2372	0.4513	0.4665	0.2844
	Std	0.0116	0.0641	0.0776	0.0624	0.0575	0.0470	0.0656	0.0651	0.0868
F1 score	Mean	0.3989	0.3585	0.2931	0.3631	0.1136	0.1485	0.3464	0.3587	0.1776
	Std	0.0099	0.0643	0.0764	0.0584	0.0515	0.0428	0.0659	0.0667	0.0828
ARI	Mean	0.5553	0.3749	0.3560	0.4026	0.1218	0.1197	0.3569	0.3742	0.2065
	Std	0.0410	0.1115	0.1092	0.1144	0.0714	0.0617	0.1081	0.1101	0.1040

Table 9.2. $N = 2000$, $V = 6$, $L = 15$, $\alpha = 0.5$, tree_depth = 4

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l^{0.5} \log(p_l)$	$b = -p_l \log(p_l)$
Accuracy	Mean	0.8796	0.6746	0.5869	0.6960	0.2942	0.2998	0.6538	0.6748	0.4119
	Std	0.0552	0.1011	0.1087	0.1113	0.1054	0.0714	0.0992	0.1004	0.1286
Precision	Mean	0.8468	0.6079	0.4903	0.6194	0.2450	0.2721	0.5972	0.6062	0.3528
	Std	0.0770	0.1133	0.1238	0.1295	0.0937	0.0935	0.1115	0.1172	0.1315
Recall	Mean	0.8796	0.6746	0.5869	0.6960	0.2942	0.2998	0.6538	0.6748	0.4119
	Std	0.0552	0.1011	0.1087	0.1113	0.1054	0.0714	0.0992	0.1004	0.1286
F1 score	Mean	0.8534	0.6027	0.4925	0.6262	0.1999	0.2200	0.5818	0.6026	0.3119
	Std	0.0702	0.1158	0.1190	0.1284	0.1040	0.0752	0.1117	0.1148	0.1343
ARI	Mean	0.8349	0.5870	0.5198	0.6158	0.1933	0.1543	0.5577	0.5893	0.3206
	Std	0.0676	0.1436	0.1472	0.1491	0.1202	0.0833	0.1406	0.1440	0.1527

Table 10.1. $N = 2000$, $V = 6$, $L = 15$, $\alpha = 0.85$, tree_depth = 3

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l^{0.5} \log(p_l)$	$b = -p_l \log(p_l)$
Accuracy	Mean	0.3658	0.3578	0.3253	0.3369	0.1676	0.1794	0.3564	0.3574	0.2539
	Std	0.0326	0.0310	0.0342	0.0336	0.0228	0.0232	0.0351	0.0317	0.0421
Precision	Mean	0.2125	0.2340	0.1794	0.1963	0.0868	0.0931	0.2290	0.2314	0.1241
	Std	0.0264	0.0278	0.0432	0.0427	0.0362	0.0355	0.0272	0.0278	0.0504
Recall	Mean	0.3658	0.3578	0.3253	0.3369	0.1676	0.1794	0.3564	0.3574	0.2539
	Std	0.0326	0.0310	0.0342	0.0336	0.0228	0.0232	0.0351	0.0317	0.0421
F1 score	Mean	0.2596	0.2612	0.2141	0.2298	0.0735	0.0937	0.2592	0.2611	0.1379
	Std	0.0280	0.0254	0.0380	0.0381	0.0201	0.0201	0.0288	0.0275	0.0419
ARI	Mean	0.2277	0.2021	0.1982	0.2016	0.0817	0.0883	0.2008	0.2018	0.1486
	Std	0.0355	0.0415	0.0358	0.0347	0.0224	0.0231	0.0426	0.0410	0.0368

Table 10.2. $N = 2000$, $V = 6$, $L = 15$, $\alpha = 0.85$, tree_depth = 4

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l^{0.5} \log(p_l)$	$b = -p_l \log(p_l)$
Accuracy	Mean	0.4739	0.4633	0.4049	0.4292	0.1888	0.1998	0.4562	0.4620	0.3079
	Std	0.0502	0.0475	0.0458	0.0457	0.0292	0.0231	0.0497	0.0483	0.0550
Precision	Mean	0.4245	0.4322	0.3116	0.3533	0.1340	0.1348	0.4302	0.4300	0.2111
	Std	0.0603	0.0570	0.0628	0.0699	0.0478	0.0349	0.0552	0.0572	0.0795
Recall	Mean	0.4739	0.4633	0.4049	0.4292	0.1888	0.1998	0.4562	0.4620	0.3079
	Std	0.0502	0.0475	0.0458	0.0457	0.0292	0.0231	0.0497	0.0483	0.0550
F1 score	Mean	0.4301	0.4200	0.3224	0.3601	0.1002	0.1209	0.4169	0.4191	0.2039
	Std	0.0543	0.0518	0.0506	0.0536	0.0275	0.0202	0.0536	0.0540	0.0632
ARI	Mean	0.2793	0.2562	0.2485	0.2540	0.0923	0.0984	0.2495	0.2543	0.1850
	Std	0.0477	0.0478	0.0440	0.0431	0.0251	0.0230	0.0446	0.0481	0.0439

Table 11.1. $N = 2000$, $V = 15$, $L = 15$, $\alpha = 0.5$, tree_depth = 3

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l^{0.5} \log(p_l)$	$b = -p_l \log(p_l)$
Accuracy	Mean	0.5469	0.4609	0.4233	0.4557	0.2424	0.2384	0.4449	0.4600	0.3077
	Std	0.0075	0.0563	0.0776	0.0611	0.0713	0.0484	0.0587	0.0598	0.0926
Precision	Mean	0.3282	0.3388	0.2824	0.3139	0.1588	0.1647	0.3469	0.3339	0.2012
	Std	0.0141	0.0504	0.0806	0.0555	0.0653	0.0542	0.0468	0.0578	0.0815
Recall	Mean	0.5469	0.4609	0.4233	0.4557	0.2424	0.2384	0.4449	0.4600	0.3077
	Std	0.0075	0.0563	0.0776	0.0611	0.0713	0.0484	0.0587	0.0598	0.0926
F1 score	Mean	0.4013	0.3599	0.3083	0.3469	0.1450	0.1537	0.3507	0.3575	0.1983
	Std	0.0096	0.0569	0.0823	0.0584	0.0683	0.0434	0.0573	0.0606	0.0915
ARI	Mean	0.6074	0.3527	0.3514	0.3720	0.1464	0.1042	0.3153	0.3572	0.2334
	Std	0.0217	0.0985	0.0962	0.1044	0.0737	0.0627	0.0903	0.0983	0.0987

Table 11.2. $N = 2000$, $V = 15$, $L = 15$, $\alpha = 0.5$, tree_depth = 4

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l^{0.5} \log(p_l)$	$b = -p_l \log(p_l)$
Accuracy	Mean	0.9474	0.6480	0.6145	0.6591	0.3253	0.2896	0.6155	0.6475	0.4464
	Std	0.0353	0.0879	0.1084	0.1006	0.1076	0.0767	0.0857	0.0917	0.1454
Precision	Mean	0.9336	0.5717	0.5038	0.5505	0.2650	0.2669	0.5543	0.5635	0.3598
	Std	0.0539	0.0894	0.1270	0.1176	0.0921	0.0881	0.0879	0.0965	0.1406
Recall	Mean	0.9474	0.6480	0.6145	0.6591	0.3253	0.2896	0.6155	0.6475	0.4464
	Std	0.0353	0.0879	0.1084	0.1006	0.1076	0.0767	0.0857	0.0917	0.1454
F1 score	Mean	0.9353	0.5748	0.5253	0.5769	0.2331	0.2159	0.5424	0.5733	0.3474
	Std	0.0473	0.0947	0.1247	0.1165	0.1062	0.0749	0.0944	0.0999	0.1545
ARI	Mean	0.9292	0.5431	0.5371	0.5968	0.2061	0.1242	0.4907	0.5482	0.3574
	Std	0.0370	0.1405	0.1347	0.1339	0.1185	0.0809	0.1188	0.1416	0.1668

Table 12.1. $N = 2000$, $V = 15$, $L = 15$, $\alpha = 0.85$, tree_depth = 3

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l^{0.5} \log(p_l)$	$b = -p_l \log(p_l)$
Accuracy	Mean	0.4146	0.4061	0.3410	0.3733	0.1486	0.1838	0.4020	0.4072	0.2076
	Std	0.0260	0.0267	0.0426	0.0441	0.0134	0.0248	0.0271	0.0269	0.0390
Precision	Mean	0.2462	0.2707	0.1825	0.2284	0.0881	0.1106	0.2697	0.2705	0.0979
	Std	0.0218	0.0234	0.0479	0.0559	0.0418	0.0374	0.0246	0.0239	0.0404
Recall	Mean	0.4146	0.4061	0.3410	0.3733	0.1486	0.1838	0.4020	0.4072	0.2076
	Std	0.0260	0.0267	0.0426	0.0441	0.0134	0.0248	0.0271	0.0269	0.0390
F1 score	Mean	0.2997	0.3041	0.2211	0.2639	0.0544	0.0982	0.3018	0.3048	0.0916
	Std	0.0227	0.0232	0.0468	0.0513	0.0137	0.0243	0.0238	0.0243	0.0313
ARI	Mean	0.2822	0.2483	0.2224	0.2334	0.0736	0.0892	0.2430	0.2489	0.1269
	Std	0.0289	0.0363	0.0384	0.0384	0.0118	0.0178	0.0386	0.0363	0.0376

Table 12.2. $N = 2000$, $V = 15$, $L = 15$, $\alpha = 0.85$, tree_depth = 4

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l^{0.5} \log(p_l)$	$b = -p_l \log(p_l)$
Accuracy	Mean	0.5874	0.5635	0.4405	0.5032	0.1560	0.2049	0.5520	0.5619	0.2442
	Std	0.0363	0.0448	0.0453	0.0556	0.0172	0.0268	0.0464	0.0440	0.0465
Precision	Mean	0.5525	0.5459	0.3252	0.4515	0.1214	0.1576	0.5347	0.5425	0.1498
	Std	0.0464	0.0561	0.0658	0.0826	0.0508	0.0496	0.0548	0.0564	0.0590
Recall	Mean	0.5874	0.5635	0.4405	0.5032	0.1560	0.2049	0.5520	0.5619	0.2442
	Std	0.0363	0.0448	0.0453	0.0556	0.0172	0.0268	0.0464	0.0440	0.0465
F1 score	Mean	0.5515	0.5263	0.3460	0.4406	0.0651	0.1257	0.5162	0.5256	0.1288
	Std	0.0389	0.0540	0.0560	0.0696	0.0158	0.0279	0.0535	0.0527	0.0448
ARI	Mean	0.3844	0.3514	0.2866	0.3110	0.0777	0.0994	0.3382	0.3498	0.1510
	Std	0.0441	0.0476	0.0407	0.0453	0.0138	0.0201	0.0499	0.0484	0.0422

Real-World UCI Data

Datasets from the UCI Machine Learning Repository used in the experiments.

No.	Dataset	N	V	L
1	Breast Cancer (Prognostic)	194	32	2
2	Balance Scale	625	4	3
3	Car Evaluation	1728	6	4
4	Connectionist Bench Sonar	208	60	2
5	Contraceptive Method Choice	1473	9	3
6	Hayes-Roth	132	4	3
7	Hepatitis	80	19	2
8	Ionosphere	351	34	2
9	Glass	214	9	6
10	Congressional Voting Records	232	16	2
11	Spambase	4601	57	2
12	Mushroom	8124	22	2
13	Yeast	1484	8	10
14	Bach Choral Harmony	5665	15	102
15	LED Display Domain	1000	7	10

Table 1.1. Breast Cancer (Prognostic), tree_depth = 3

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l^{0.5} \log(p_l)$	$b = -p_l \log(p_l)$
Accuracy	Mean	0.7098	0.6955	0.6959	0.6955	0.6951	0.6959	0.6963	0.6959	0.6959
	Std	0.0870	0.0731	0.0728	0.0723	0.0727	0.0730	0.0718	0.0727	0.0735
Precision	Mean	0.6490	0.6558	0.6562	0.6561	0.6561	0.6564	0.6564	0.6547	0.6548
	Std	0.0890	0.0874	0.0872	0.0870	0.0871	0.0874	0.0868	0.0874	0.0877
Recall	Mean	0.7098	0.6955	0.6959	0.6955	0.6951	0.6959	0.6963	0.6959	0.6959
	Std	0.0870	0.0731	0.0728	0.0723	0.0727	0.0730	0.0718	0.0727	0.0735
F1 score	Mean	0.6654	0.6646	0.6650	0.6647	0.6645	0.6651	0.6652	0.6646	0.6646
	Std	0.0777	0.0783	0.0783	0.0778	0.0781	0.0786	0.0774	0.0778	0.0783
ARI	Mean	0.0076	0.0175	0.0180	0.0174	0.0174	0.0182	0.0178	0.0176	0.0181
	Std	0.0699	0.0887	0.0909	0.0888	0.0890	0.0915	0.0884	0.0905	0.0905

Table 1.2. Breast Cancer (Prognostic), tree_depth = 4

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l^{0.5} \log(p_l)$	$b = -p_l \log(p_l)$
Accuracy	Mean	0.6882	0.6629	0.6620	0.6625	0.6629	0.6620	0.6625	0.6616	0.6637
	Std	0.0832	0.0639	0.0648	0.0657	0.0631	0.0658	0.0655	0.0657	0.0671
Precision	Mean	0.6598	0.6516	0.6515	0.6517	0.6522	0.6516	0.6512	0.6505	0.6516
	Std	0.0859	0.0800	0.0781	0.0780	0.0783	0.0784	0.0781	0.0791	0.0782
Recall	Mean	0.6882	0.6629	0.6620	0.6625	0.6629	0.6620	0.6625	0.6616	0.6637
	Std	0.0832	0.0639	0.0648	0.0657	0.0631	0.0658	0.0655	0.0657	0.0671
F1 score	Mean	0.6616	0.6499	0.6496	0.6500	0.6504	0.6498	0.6500	0.6490	0.6506
	Std	0.0831	0.0696	0.0700	0.0698	0.0686	0.0701	0.0698	0.0703	0.0708
ARI	Mean	0.0238	0.0079	0.0071	0.0073	0.0079	0.0073	0.0073	0.0065	0.0083
	Std	0.0708	0.0650	0.0618	0.0621	0.0632	0.0626	0.0619	0.0624	0.0635

Table 2.1. Balance Scale, tree_depth = 3

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l^{0.5} \log(p_l)$	$b = -p_l \log(p_l)$
Accuracy	Mean	0.6989	0.7092	0.7085	0.7085	0.7084	0.6912	0.7057	0.7062	0.7062
	Std	0.0266	0.0326	0.0287	0.0310	0.0280	0.0444	0.0316	0.0304	0.0307
Precision	Mean	0.6511	0.6612	0.6617	0.6600	0.6601	0.6498	0.6576	0.6573	0.6576
	Std	0.0316	0.0370	0.0334	0.0355	0.0329	0.0451	0.0347	0.0338	0.0344
Recall	Mean	0.6989	0.7092	0.7085	0.7085	0.7084	0.6912	0.7057	0.7062	0.7062
	Std	0.0266	0.0326	0.0287	0.0310	0.0280	0.0444	0.0316	0.0304	0.0307
F1 score	Mean	0.6707	0.6811	0.6803	0.6804	0.6800	0.6611	0.6774	0.6781	0.6782
	Std	0.0292	0.0344	0.0300	0.0326	0.0304	0.0471	0.0331	0.0319	0.0323
ARI	Mean	0.2219	0.2427	0.2406	0.2414	0.2403	0.2129	0.2366	0.2374	0.2372
	Std	0.0457	0.0606	0.0534	0.0568	0.0505	0.0763	0.0588	0.0571	0.0574

Table 2.2. Balance Scale, tree_depth = 4

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l^{0.5} \log(p_l)$	$b = -p_l \log(p_l)$
Accuracy	Mean	0.7442	0.7825	0.7757	0.7782	0.7526	0.7027	0.7731	0.7766	0.7721
	Std	0.0261	0.0320	0.0293	0.0310	0.0323	0.0523	0.0293	0.0310	0.0318
Precision	Mean	0.7067	0.7334	0.7277	0.7295	0.7105	0.6758	0.7290	0.7322	0.7251
	Std	0.0326	0.0357	0.0305	0.0328	0.0334	0.0485	0.0385	0.0397	0.0345
Recall	Mean	0.7442	0.7825	0.7757	0.7782	0.7526	0.7027	0.7731	0.7766	0.7721
	Std	0.0261	0.0320	0.0293	0.0310	0.0323	0.0523	0.0293	0.0310	0.0318
F1 score	Mean	0.7199	0.7551	0.7483	0.7508	0.7256	0.6748	0.7464	0.7499	0.7449
	Std	0.0280	0.0332	0.0290	0.0310	0.0324	0.0538	0.0312	0.0326	0.0330
ARI	Mean	0.3239	0.4173	0.3993	0.4063	0.3417	0.2400	0.3922	0.4016	0.3910
	Std	0.0633	0.0757	0.0724	0.0744	0.0760	0.0961	0.0690	0.0731	0.0758

Table 3.1. Car Evaluation, tree_depth = 3

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l^{0.5} \log(p_l)$	$b = -p_l \log(p_l)$
Accuracy	Mean	0.7887	0.7878	0.7887	0.7887	0.7656	0.7163	0.7898	0.7879	0.7887
	Std	0.0190	0.0190	0.0190	0.0190	0.0214	0.0249	0.0146	0.0166	0.0190
Precision	Mean	0.7300	0.7300	0.7300	0.7300	0.7637	0.6665	0.7448	0.7302	0.7300
	Std	0.0271	0.0271	0.0271	0.0271	0.0560	0.0505	0.0290	0.0257	0.0271
Recall	Mean	0.7887	0.7878	0.7887	0.7887	0.7656	0.7163	0.7898	0.7879	0.7887
	Std	0.0190	0.0190	0.0190	0.0190	0.0214	0.0249	0.0146	0.0166	0.0190
F1 score	Mean	0.7554	0.7549	0.7554	0.7554	0.7483	0.6784	0.7627	0.7554	0.7554
	Std	0.0172	0.0170	0.0172	0.0172	0.0248	0.0411	0.0185	0.0158	0.0172
ARI	Mean	0.5088	0.5061	0.5088	0.5088	0.4003	0.2121	0.5012	0.5058	0.5088
	Std	0.0527	0.0530	0.0527	0.0527	0.0514	0.0695	0.0393	0.0480	0.0527

Table 3.2. Car Evaluation, tree_depth = 4

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l^{0.5} \log(p_l)$	$b = -p_l \log(p_l)$
Accuracy	Mean	0.8339	0.8443	0.8443	0.8443	0.8310	0.7467	0.8366	0.8441	0.8443
	Std	0.0183	0.0151	0.0151	0.0151	0.0242	0.0219	0.0197	0.0167	0.0151
Precision	Mean	0.8523	0.8451	0.8458	0.8458	0.7947	0.7268	0.8384	0.8458	0.8458
	Std	0.0207	0.0223	0.0222	0.0222	0.0297	0.0305	0.0203	0.0203	0.0222
Recall	Mean	0.8339	0.8443	0.8443	0.8443	0.8310	0.7467	0.8366	0.8441	0.8443
	Std	0.0183	0.0151	0.0151	0.0151	0.0242	0.0219	0.0197	0.0167	0.0151
F1 score	Mean	0.8308	0.8371	0.8376	0.8376	0.8091	0.7300	0.8256	0.8369	0.8376
	Std	0.0198	0.0186	0.0181	0.0181	0.0266	0.0254	0.0235	0.0195	0.0181
ARI	Mean	0.6506	0.6821	0.6836	0.6836	0.6187	0.3225	0.6364	0.6762	0.6836
	Std	0.0554	0.0489	0.0464	0.0464	0.0591	0.0513	0.0724	0.0603	0.0464

Table 4.1. Connectionist Bench Sonar, tree_depth = 3

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l^{0.5} \log(p_l)$	$b = -p_l \log(p_l)$
Accuracy	Mean	0.7238	0.7162	0.7138	0.7165	0.7127	0.7138	0.7138	0.7146	0.7162
	Std	0.0660	0.0671	0.0667	0.0667	0.0661	0.0661	0.0661	0.0670	0.0670
Precision	Mean	0.7475	0.7292	0.7271	0.7300	0.7266	0.7274	0.7274	0.7282	0.7296
	Std	0.0703	0.0703	0.0690	0.0697	0.0687	0.0690	0.0690	0.0699	0.0697
Recall	Mean	0.7238	0.7162	0.7138	0.7165	0.7127	0.7138	0.7138	0.7146	0.7162
	Std	0.0660	0.0671	0.0667	0.0667	0.0661	0.0661	0.0661	0.0670	0.0670
F1 score	Mean	0.7192	0.7133	0.7110	0.7136	0.7096	0.7108	0.7108	0.7116	0.7130
	Std	0.0676	0.0688	0.0680	0.0685	0.0678	0.0679	0.0679	0.0687	0.0688
ARI	Mean	0.2019	0.1892	0.1850	0.1897	0.1826	0.1846	0.1846	0.1864	0.1891
	Std	0.1162	0.1148	0.1131	0.1141	0.1127	0.1125	0.1125	0.1146	0.1139

Table 4.2. Connectionist Bench Sonar, tree_depth = 4

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l^{0.5} \log(p_l)$	$b = -p_l \log(p_l)$
Accuracy	Mean	0.7235	0.7131	0.7138	0.7115	0.7092	0.7123	0.7104	0.7119	0.7115
	Std	0.0640	0.0655	0.0653	0.0658	0.0628	0.0641	0.0643	0.0646	0.0664
Precision	Mean	0.7372	0.7229	0.7233	0.7215	0.7193	0.7219	0.7197	0.7217	0.7207
	Std	0.0623	0.0655	0.0646	0.0663	0.0622	0.0639	0.0644	0.0646	0.0661
Recall	Mean	0.7235	0.7131	0.7138	0.7115	0.7092	0.7123	0.7104	0.7119	0.7115
	Std	0.0640	0.0655	0.0653	0.0658	0.0628	0.0641	0.0643	0.0646	0.0664
F1 score	Mean	0.7219	0.7118	0.7125	0.7103	0.7077	0.7109	0.7089	0.7105	0.7102
	Std	0.0640	0.0667	0.0667	0.0669	0.0643	0.0656	0.0658	0.0660	0.0676
ARI	Mean	0.2005	0.1830	0.1842	0.1804	0.1749	0.1809	0.1776	0.1805	0.1807
	Std	0.1120	0.1125	0.1153	0.1142	0.1071	0.1113	0.1119	0.1125	0.1163

Table 5.1. Contraceptive Method Choice, tree_depth = 3

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l^{0.5} \log(p_l)$	$b = -p_l \log(p_l)$
Accuracy	Mean	0.5018	0.5118	0.5150	0.5131	0.5125	0.4840	0.5130	0.5120	0.5122
	Std	0.0291	0.0269	0.0282	0.0279	0.0291	0.0236	0.0311	0.0276	0.0267
Precision	Mean	0.5463	0.5677	0.5709	0.5754	0.5418	0.4450	0.5160	0.5469	0.5728
	Std	0.0771	0.0564	0.0578	0.0524	0.0699	0.0737	0.0666	0.0669	0.0553
Recall	Mean	0.5018	0.5118	0.5150	0.5131	0.5125	0.4840	0.5130	0.5120	0.5122
	Std	0.0291	0.0269	0.0282	0.0279	0.0291	0.0236	0.0311	0.0276	0.0267
F1 score	Mean	0.4857	0.4992	0.5077	0.5052	0.5040	0.4535	0.4884	0.4921	0.5015
	Std	0.0427	0.0317	0.0381	0.0348	0.0469	0.0526	0.0426	0.0374	0.0321
ARI	Mean	0.0606	0.0676	0.0735	0.0698	0.0776	0.0758	0.0727	0.0679	0.0677
	Std	0.0246	0.0229	0.0248	0.0244	0.0223	0.0184	0.0270	0.0262	0.0232

Table 5.2. Contraceptive Method Choice, tree_depth = 4

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l^{0.5} \log(p_l)$	$b = -p_l \log(p_l)$
Accuracy	Mean	0.5436	0.5512	0.5499	0.5497	0.5315	0.5041	0.5382	0.5458	0.5507
	Std	0.0260	0.0237	0.0221	0.0227	0.0218	0.0277	0.0246	0.0240	0.0231
Precision	Mean	0.5623	0.5648	0.5566	0.5618	0.5584	0.4960	0.5597	0.5616	0.5627
	Std	0.0368	0.0284	0.0235	0.0260	0.0360	0.0336	0.0296	0.0293	0.0251
Recall	Mean	0.5436	0.5512	0.5499	0.5497	0.5315	0.5041	0.5382	0.5458	0.5507
	Std	0.0260	0.0237	0.0221	0.0227	0.0218	0.0277	0.0246	0.0240	0.0231
F1 score	Mean	0.5374	0.5455	0.5457	0.5450	0.5283	0.4876	0.5313	0.5394	0.5463
	Std	0.0324	0.0244	0.0240	0.0242	0.0275	0.0368	0.0242	0.0249	0.0237
ARI	Mean	0.1026	0.1115	0.1137	0.1111	0.0935	0.0887	0.0965	0.1067	0.1119
	Std	0.0302	0.0266	0.0261	0.0260	0.0234	0.0218	0.0271	0.0278	0.0256

Table 6.1. Hayes-Roth, tree_depth = 3

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l^{0.5} \log(p_l)$	$b = -p_l \log(p_l)$
Accuracy	Mean	0.5528	0.5503	0.5528	0.5528	0.5528	0.5528	0.5376	0.5443	0.5528
	Std	0.0602	0.0628	0.0602	0.0602	0.0602	0.0602	0.0517	0.0549	0.0602
Precision	Mean	0.3704	0.3730	0.3704	0.3704	0.3704	0.3704	0.5587	0.3946	0.3704
	Std	0.0585	0.0605	0.0585	0.0585	0.0585	0.0585	0.1277	0.0893	0.0585
Recall	Mean	0.5528	0.5503	0.5528	0.5528	0.5528	0.5528	0.5376	0.5443	0.5528
	Std	0.0602	0.0628	0.0602	0.0602	0.0602	0.0602	0.0517	0.0549	0.0602
F1 score	Mean	0.4241	0.4243	0.4241	0.4241	0.4241	0.4241	0.5179	0.4316	0.4241
	Std	0.0630	0.0630	0.0630	0.0630	0.0630	0.0630	0.0894	0.0607	0.0630
ARI	Mean	0.4657	0.4584	0.4657	0.4657	0.4657	0.4657	0.1861	0.4163	0.4657
	Std	0.0698	0.0893	0.0698	0.0698	0.0698	0.0698	0.1467	0.1445	0.0698

Table 6.2. Hayes-Roth, tree_depth = 4

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l^{0.5} \log(p_l)$	$b = -p_l \log(p_l)$
Accuracy	Mean	0.6564	0.6564	0.6564	0.6564	0.6564	0.6564	0.6115	0.6455	0.6564
	Std	0.0448	0.0448	0.0448	0.0448	0.0448	0.0448	0.0909	0.0636	0.0448
Precision	Mean	0.6759	0.6759	0.6759	0.6759	0.6759	0.6759	0.6764	0.6696	0.6759
	Std	0.0625	0.0625	0.0625	0.0625	0.0625	0.0625	0.0814	0.0663	0.0625
Recall	Mean	0.6564	0.6564	0.6564	0.6564	0.6564	0.6564	0.6115	0.6455	0.6564
	Std	0.0448	0.0448	0.0448	0.0448	0.0448	0.0448	0.0909	0.0636	0.0448
F1 score	Mean	0.6551	0.6551	0.6551	0.6551	0.6551	0.6551	0.6059	0.6443	0.6551
	Std	0.0476	0.0476	0.0476	0.0476	0.0476	0.0476	0.0943	0.0651	0.0476
ARI	Mean	0.3449	0.3449	0.3449	0.3449	0.3449	0.3449	0.2542	0.3300	0.3449
	Std	0.0757	0.0757	0.0757	0.0757	0.0757	0.0757	0.1144	0.0874	0.0757

Table 7.1. Hepatitis, tree_depth = 3

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l^{0.5} \log(p_l)$	$b = -p_l \log(p_l)$
Accuracy	Mean	0.8170	0.8110	0.8080	0.8080	0.8080	0.8060	0.8120	0.8080	0.8090
	Std	0.0704	0.0666	0.0681	0.0681	0.0681	0.0668	0.0690	0.0681	0.0691
Precision	Mean	0.8340	0.8166	0.8137	0.8137	0.8134	0.8136	0.8198	0.8137	0.8152
	Std	0.0949	0.0977	0.1001	0.1001	0.1000	0.1000	0.0994	0.1001	0.0994
Recall	Mean	0.8170	0.8110	0.8080	0.8080	0.8080	0.8060	0.8120	0.8080	0.8090
	Std	0.0704	0.0666	0.0681	0.0681	0.0681	0.0668	0.0690	0.0681	0.0691
F1 score	Mean	0.8122	0.8071	0.8038	0.8038	0.8036	0.8027	0.8087	0.8038	0.8053
	Std	0.0810	0.0783	0.0805	0.0805	0.0806	0.0796	0.0795	0.0805	0.0804
ARI	Mean	0.2194	0.1992	0.1937	0.1937	0.1915	0.1934	0.2084	0.1937	0.1986
	Std	0.2126	0.1894	0.1905	0.1905	0.1890	0.1910	0.1978	0.1905	0.1973

Table 7.2. Hepatitis, tree_depth = 4

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l^{0.5} \log(p_l)$	$b = -p_l \log(p_l)$
Accuracy	Mean	0.8180	0.8100	0.8050	0.8060	0.8070	0.8080	0.8100	0.8070	0.8040
	Std	0.0705	0.0686	0.0709	0.0712	0.0700	0.0724	0.0721	0.0707	0.0720
Precision	Mean	0.8373	0.8280	0.8214	0.8233	0.8248	0.8252	0.8307	0.8251	0.8224
	Std	0.0962	0.0919	0.0979	0.0961	0.0947	0.0948	0.0935	0.0948	0.0972
Recall	Mean	0.8180	0.8100	0.8050	0.8060	0.8070	0.8080	0.8100	0.8070	0.8040
	Std	0.0705	0.0686	0.0709	0.0712	0.0700	0.0724	0.0721	0.0707	0.0720
F1 score	Mean	0.8141	0.8129	0.8068	0.8080	0.8093	0.8100	0.8136	0.8094	0.8069
	Std	0.0827	0.0763	0.0804	0.0797	0.0789	0.0796	0.0778	0.0789	0.0803
ARI	Mean	0.2288	0.2294	0.2151	0.2178	0.2217	0.2238	0.2356	0.2228	0.2153
	Std	0.2173	0.1911	0.1956	0.1987	0.1921	0.1921	0.1989	0.1935	0.1950

Table 8.1. Ionosphere, tree_depth = 3

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l^{0.5} \log(p_l)$	$b = -p_l \log(p_l)$
Accuracy	Mean	0.8984	0.8866	0.8868	0.8868	0.8866	0.8866	0.8866	0.8866	0.8866
	Std	0.0372	0.0418	0.0415	0.0415	0.0418	0.0418	0.0418	0.0418	0.0416
Precision	Mean	0.9028	0.8924	0.8927	0.8927	0.8924	0.8924	0.8924	0.8924	0.8924
	Std	0.0312	0.0368	0.0362	0.0362	0.0368	0.0368	0.0368	0.0368	0.0368
Recall	Mean	0.8984	0.8866	0.8868	0.8868	0.8866	0.8866	0.8866	0.8866	0.8866
	Std	0.0372	0.0418	0.0415	0.0415	0.0418	0.0418	0.0418	0.0418	0.0416
F1 score	Mean	0.8978	0.8857	0.8859	0.8859	0.8857	0.8857	0.8857	0.8857	0.8857
	Std	0.0395	0.0445	0.0442	0.0442	0.0445	0.0445	0.0445	0.0445	0.0444
ARI	Mean	0.6325	0.5963	0.5970	0.5970	0.5963	0.5963	0.5963	0.5963	0.5963
	Std	0.1144	0.1265	0.1255	0.1255	0.1265	0.1265	0.1265	0.1265	0.1258

Table 8.2. Ionosphere, tree_depth = 4

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l^{0.5} \log(p_l)$	$b = -p_l \log(p_l)$
Accuracy	Mean	0.8761	0.8820	0.8825	0.8820	0.8809	0.8818	0.8820	0.8816	0.8845
	Std	0.0362	0.0322	0.0306	0.0318	0.0327	0.0318	0.0322	0.0322	0.0321
Precision	Mean	0.8809	0.8860	0.8864	0.8862	0.8850	0.8857	0.8860	0.8854	0.8886
	Std	0.0366	0.0323	0.0306	0.0316	0.0325	0.0318	0.0323	0.0323	0.0318
Recall	Mean	0.8761	0.8820	0.8825	0.8820	0.8809	0.8818	0.8820	0.8816	0.8845
	Std	0.0362	0.0322	0.0306	0.0318	0.0327	0.0318	0.0322	0.0322	0.0321
F1 score	Mean	0.8731	0.8795	0.8801	0.8795	0.8784	0.8793	0.8795	0.8791	0.8820
	Std	0.0373	0.0335	0.0318	0.0330	0.0338	0.0331	0.0335	0.0334	0.0332
ARI	Mean	0.5581	0.5757	0.5769	0.5756	0.5724	0.5749	0.5757	0.5743	0.5836
	Std	0.1108	0.1016	0.0975	0.0998	0.1028	0.1005	0.1016	0.1016	0.1021

Table 9.1. Glass, tree_depth = 3

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l^{0.5} \log(p_l)$	$b = -p_l \log(p_l)$
Accuracy	Mean	0.6318	0.6533	0.5456	0.5711	0.4363	0.4144	0.6207	0.6415	0.5074
	Std	0.0567	0.0512	0.0953	0.1033	0.0819	0.0557	0.0561	0.0655	0.0933
Precision	Mean	0.6351	0.6040	0.4600	0.5195	0.3107	0.2822	0.6075	0.6059	0.3872
	Std	0.0684	0.0646	0.1688	0.1792	0.1412	0.1363	0.0703	0.0731	0.1525
Recall	Mean	0.6318	0.6533	0.5456	0.5711	0.4363	0.4144	0.6207	0.6415	0.5074
	Std	0.0567	0.0512	0.0953	0.1033	0.0819	0.0557	0.0561	0.0655	0.0933
F1 score	Mean	0.6141	0.6134	0.4603	0.5028	0.3177	0.2847	0.5952	0.6076	0.4139
	Std	0.0593	0.0517	0.1334	0.1435	0.1002	0.0688	0.0573	0.0644	0.1302
ARI	Mean	0.2926	0.3154	0.3067	0.3092	0.2345	0.2257	0.2857	0.3003	0.2657
	Std	0.0759	0.0749	0.0712	0.0725	0.0646	0.0562	0.0747	0.0849	0.0560

Table 9.2. Glass, tree_depth = 4

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l^{0.5} \log(p_l)$	$b = -p_l \log(p_l)$
Accuracy	Mean	0.6637	0.6578	0.5952	0.6393	0.5081	0.4474	0.6381	0.6537	0.5870
	Std	0.0585	0.0569	0.1039	0.0763	0.1200	0.0771	0.0579	0.0702	0.1098
Precision	Mean	0.6804	0.6515	0.5613	0.6286	0.4577	0.3928	0.6511	0.6596	0.5392
	Std	0.0578	0.0790	0.1648	0.1033	0.1731	0.1742	0.0621	0.0862	0.1719
Recall	Mean	0.6637	0.6578	0.5952	0.6393	0.5081	0.4474	0.6381	0.6537	0.5870
	Std	0.0585	0.0569	0.1039	0.0763	0.1200	0.0771	0.0579	0.0702	0.1098
F1 score	Mean	0.6547	0.6386	0.5462	0.6138	0.4273	0.3389	0.6262	0.6381	0.5365
	Std	0.0608	0.0632	0.1429	0.0926	0.1549	0.1020	0.0590	0.0767	0.1518
ARI	Mean	0.3185	0.3119	0.2975	0.3047	0.2673	0.2400	0.2881	0.3047	0.2888
	Std	0.0782	0.0825	0.0739	0.0763	0.0798	0.0605	0.0729	0.0927	0.0804

Table 10.1. Congressional Voting Records, tree_depth = 3

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l^{0.5} \log(p_l)$	$b = -p_l \log(p_l)$
Accuracy	Mean	0.9507	0.9493	0.9493	0.9493	0.9493	0.9493	0.9486	0.9493	0.9493
	Std	0.0272	0.0270	0.0270	0.0270	0.0270	0.0270	0.0266	0.0270	0.0270
Precision	Mean	0.9534	0.9524	0.9524	0.9524	0.9524	0.9524	0.9517	0.9524	0.9524
	Std	0.0244	0.0241	0.0241	0.0241	0.0241	0.0241	0.0237	0.0241	0.0241
Recall	Mean	0.9507	0.9493	0.9493	0.9493	0.9493	0.9493	0.9486	0.9493	0.9493
	Std	0.0272	0.0270	0.0270	0.0270	0.0270	0.0270	0.0266	0.0270	0.0270
F1 score	Mean	0.9506	0.9493	0.9493	0.9493	0.9493	0.9493	0.9486	0.9493	0.9493
	Std	0.0272	0.0271	0.0271	0.0271	0.0271	0.0271	0.0267	0.0271	0.0271
ARI	Mean	0.8121	0.8070	0.8070	0.8070	0.8070	0.8070	0.8044	0.8070	0.8070
	Std	0.0973	0.0965	0.0965	0.0965	0.0965	0.0965	0.0948	0.0965	0.0965

Table 10.2. Congressional Voting Records, tree_depth = 4

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l^{0.5} \log(p_l)$	$b = -p_l \log(p_l)$
Accuracy	Mean	0.9517	0.9483	0.9497	0.9493	0.9483	0.9486	0.9490	0.9483	0.9490
	Std	0.0282	0.0272	0.0271	0.0272	0.0276	0.0275	0.0280	0.0278	0.0271
Precision	Mean	0.9544	0.9513	0.9525	0.9521	0.9513	0.9517	0.9519	0.9512	0.9520
	Std	0.0244	0.0241	0.0242	0.0243	0.0246	0.0244	0.0251	0.0248	0.0239
Recall	Mean	0.9517	0.9483	0.9497	0.9493	0.9483	0.9486	0.9490	0.9483	0.9490
	Std	0.0282	0.0272	0.0271	0.0272	0.0276	0.0275	0.0280	0.0278	0.0271
F1 score	Mean	0.9517	0.9483	0.9496	0.9493	0.9483	0.9486	0.9490	0.9483	0.9489
	Std	0.0283	0.0272	0.0272	0.0273	0.0276	0.0275	0.0281	0.0279	0.0272
ARI	Mean	0.8162	0.8033	0.8083	0.8071	0.8034	0.8046	0.8060	0.8035	0.8058
	Std	0.1011	0.0969	0.0970	0.0975	0.0987	0.0982	0.1002	0.0994	0.0974

Table 11.1. Spambase, tree_depth = 3

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l^{0.5} \log(p_l)$	$b = -p_l \log(p_l)$
Accuracy	Mean	0.8693	0.8820	0.8820	0.8820	0.8820	0.8820	0.8820	0.8820	0.8820
	Std	0.0086	0.0111	0.0111	0.0111	0.0111	0.0111	0.0111	0.0111	0.0111
Precision	Mean	0.8757	0.8832	0.8832	0.8832	0.8832	0.8832	0.8832	0.8832	0.8832
	Std	0.0085	0.0105	0.0105	0.0105	0.0105	0.0105	0.0105	0.0105	0.0105
Recall	Mean	0.8693	0.8820	0.8820	0.8820	0.8820	0.8820	0.8820	0.8820	0.8820
	Std	0.0086	0.0111	0.0111	0.0111	0.0111	0.0111	0.0111	0.0111	0.0111
F1 score	Mean	0.8666	0.8810	0.8810	0.8810	0.8810	0.8810	0.8810	0.8810	0.8810
	Std	0.0094	0.0113	0.0113	0.0113	0.0113	0.0113	0.0113	0.0113	0.0113
ARI	Mean	0.5429	0.5824	0.5824	0.5824	0.5824	0.5824	0.5824	0.5824	0.5824
	Std	0.0258	0.0337	0.0337	0.0337	0.0337	0.0337	0.0337	0.0337	0.0337

Table 11.2. Spambase, tree_depth = 4

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l^{0.5} \log(p_l)$	$b = -p_l \log(p_l)$
Accuracy	Mean	0.8982	0.8976	0.8976	0.8976	0.8976	0.8976	0.8976	0.8976	0.8976
	Std	0.0086	0.0104	0.0104	0.0104	0.0104	0.0104	0.0104	0.0104	0.0104
Precision	Mean	0.8997	0.8995	0.8995	0.8995	0.8995	0.8995	0.8995	0.8995	0.8995
	Std	0.0084	0.0105	0.0105	0.0105	0.0105	0.0105	0.0105	0.0105	0.0105
Recall	Mean	0.8982	0.8976	0.8976	0.8976	0.8976	0.8976	0.8976	0.8976	0.8976
	Std	0.0086	0.0104	0.0104	0.0104	0.0104	0.0104	0.0104	0.0104	0.0104
F1 score	Mean	0.8969	0.8964	0.8964	0.8964	0.8964	0.8964	0.8964	0.8964	0.8964
	Std	0.0087	0.0104	0.0104	0.0104	0.0104	0.0104	0.0104	0.0104	0.0104
ARI	Mean	0.6326	0.6310	0.6310	0.6309	0.6310	0.6310	0.6310	0.6310	0.6310
	Std	0.0271	0.0330	0.0330	0.0330	0.0330	0.0330	0.0330	0.0330	0.0330

Table 12.1. Mushroom, tree_depth = 3

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l^{0.5} \log(p_l)$	$b = -p_l \log(p_l)$
Accuracy	Mean	0.9568	0.9588	0.9588	0.9588	0.9588	0.9588	0.9588	0.9588	0.9588
	Std	0.0036	0.0041	0.0041	0.0041	0.0041	0.0041	0.0041	0.0041	0.0041
Precision	Mean	0.9574	0.9592	0.9592	0.9592	0.9592	0.9592	0.9592	0.9592	0.9592
	Std	0.0035	0.0040	0.0040	0.0040	0.0040	0.0040	0.0040	0.0040	0.0040
Recall	Mean	0.9568	0.9588	0.9588	0.9588	0.9588	0.9588	0.9588	0.9588	0.9588
	Std	0.0036	0.0041	0.0041	0.0041	0.0041	0.0041	0.0041	0.0041	0.0041
F1 score	Mean	0.9568	0.9588	0.9588	0.9588	0.9588	0.9588	0.9588	0.9588	0.9588
	Std	0.0036	0.0041	0.0041	0.0041	0.0041	0.0041	0.0041	0.0041	0.0041
ARI	Mean	0.8346	0.8419	0.8419	0.8419	0.8419	0.8419	0.8419	0.8419	0.8419
	Std	0.0133	0.0149	0.0149	0.0149	0.0149	0.0149	0.0149	0.0149	0.0149

Table 12.2. Mushroom, tree_depth = 4

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l^{0.5} \log(p_l)$	$b = -p_l \log(p_l)$
Accuracy	Mean	0.9582	0.9774	0.9774	0.9774	0.9774	0.9774	0.9774	0.9774	0.9774
	Std	0.0059	0.0038	0.0038	0.0038	0.0038	0.0038	0.0038	0.0038	0.0038
Precision	Mean	0.9601	0.9777	0.9777	0.9777	0.9777	0.9777	0.9777	0.9777	0.9777
	Std	0.0053	0.0035	0.0035	0.0035	0.0035	0.0035	0.0035	0.0035	0.0035
Recall	Mean	0.9582	0.9774	0.9774	0.9774	0.9774	0.9774	0.9774	0.9774	0.9774
	Std	0.0059	0.0038	0.0038	0.0038	0.0038	0.0038	0.0038	0.0038	0.0038
F1 score	Mean	0.9582	0.9774	0.9774	0.9774	0.9774	0.9774	0.9774	0.9774	0.9774
	Std	0.0059	0.0038	0.0038	0.0038	0.0038	0.0038	0.0038	0.0038	0.0038
ARI	Mean	0.8399	0.9117	0.9117	0.9117	0.9117	0.9117	0.9117	0.9117	0.9117
	Std	0.0216	0.0142	0.0142	0.0142	0.0142	0.0142	0.0142	0.0142	0.0142

Table 13.1. Yeast, tree_depth = 3

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l^{0.5} \log(p_l)$	$b = -p_l \log(p_l)$
Accuracy	Mean	0.5035	0.5598	0.4387	0.4481	0.3273	0.3405	0.5521	0.5552	0.3345
	Std	0.0268	0.0204	0.0279	0.0246	0.0248	0.0378	0.0204	0.0201	0.0304
Precision	Mean	0.3637	0.5488	0.2440	0.2620	0.1372	0.1656	0.5441	0.5453	0.1600
	Std	0.0440	0.0250	0.0476	0.0620	0.0462	0.0787	0.0220	0.0242	0.0543
Recall	Mean	0.5035	0.5598	0.4387	0.4481	0.3273	0.3405	0.5521	0.5552	0.3345
	Std	0.0268	0.0204	0.0279	0.0246	0.0248	0.0378	0.0204	0.0201	0.0304
F1 score	Mean	0.4093	0.5344	0.2950	0.3042	0.1712	0.1995	0.5281	0.5299	0.1809
	Std	0.0358	0.0232	0.0279	0.0262	0.0214	0.0587	0.0221	0.0226	0.0280
ARI	Mean	0.2700	0.2148	0.2244	0.2259	0.0879	0.0985	0.2089	0.2108	0.0949
	Std	0.0260	0.0214	0.0238	0.0217	0.0161	0.0177	0.0201	0.0207	0.0240

Table 13.2. Yeast, tree_depth = 4

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l^{0.5} \log(p_l)$	$b = -p_l \log(p_l)$
Accuracy	Mean	0.5691	0.5650	0.4783	0.4720	0.3340	0.3730	0.5621	0.5640	0.3802
	Std	0.0207	0.0204	0.0319	0.0292	0.0336	0.0482	0.0214	0.0209	0.0464
Precision	Mean	0.5631	0.5653	0.3570	0.3619	0.1840	0.2540	0.5635	0.5656	0.2285
	Std	0.0226	0.0247	0.0772	0.0709	0.0663	0.0923	0.0242	0.0234	0.0744
Recall	Mean	0.5691	0.5650	0.4783	0.4720	0.3340	0.3730	0.5621	0.5640	0.3802
	Std	0.0207	0.0204	0.0319	0.0292	0.0336	0.0482	0.0214	0.0209	0.0464
F1 score	Mean	0.5520	0.5461	0.3639	0.3485	0.1836	0.2684	0.5423	0.5464	0.2330
	Std	0.0231	0.0227	0.0450	0.0410	0.0322	0.0755	0.0234	0.0231	0.0479
ARI	Mean	0.2445	0.2365	0.2567	0.2424	0.1017	0.1161	0.2340	0.2360	0.1437
	Std	0.0224	0.0226	0.0332	0.0260	0.0302	0.0402	0.0212	0.0224	0.0580

Table 14.1. Bach Choral Harmony, tree_depth = 3

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l^{0.5} \log(p_l)$	$b = -p_l \log(p_l)$
Accuracy	Mean	0.4086	0.4162	0.2412	0.2711	0.1524	0.2876	0.4146	0.4054	0.1692
	Std	0.0233	0.0115	0.0108	0.0364	0.0226	0.0300	0.0140	0.0139	0.0219
Precision	Mean	0.2463	0.2659	0.0844	0.1152	0.0405	0.1191	0.2693	0.2569	0.0522
	Std	0.0211	0.0132	0.0109	0.0272	0.0118	0.0262	0.0153	0.0168	0.0185
Recall	Mean	0.4086	0.4162	0.2412	0.2711	0.1524	0.2876	0.4146	0.4054	0.1692
	Std	0.0233	0.0115	0.0108	0.0364	0.0226	0.0300	0.0140	0.0139	0.0219
F1 score	Mean	0.2930	0.3098	0.1146	0.1486	0.0573	0.1618	0.3119	0.2990	0.0700
	Std	0.0236	0.0122	0.0088	0.0361	0.0150	0.0309	0.0145	0.0164	0.0162
ARI	Mean	0.2690	0.2675	0.1145	0.1327	0.0416	0.1521	0.2564	0.2467	0.0525
	Std	0.0246	0.0163	0.0122	0.0306	0.0169	0.0173	0.0155	0.0240	0.0128

Table 14.2. Bach Choral Harmony, tree_depth = 4

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l^{0.5} \log(p_l)$	$b = -p_l \log(p_l)$
Accuracy	Mean	0.5416	0.5422	0.3168	0.3752	0.1960	0.3434	0.5281	0.5260	0.2114
	Std	0.0186	0.0121	0.0205	0.0574	0.0273	0.0363	0.0186	0.0183	0.0152
Precision	Mean	0.4174	0.4388	0.1566	0.2226	0.0653	0.1798	0.4285	0.4159	0.0774
	Std	0.0261	0.0186	0.0295	0.0624	0.0188	0.0345	0.0290	0.0278	0.0179
Recall	Mean	0.5416	0.5422	0.3168	0.3752	0.1960	0.3434	0.5281	0.5260	0.2114
	Std	0.0186	0.0121	0.0205	0.0574	0.0273	0.0363	0.0186	0.0183	0.0152
F1 score	Mean	0.4586	0.4687	0.1891	0.2608	0.0865	0.2214	0.4566	0.4474	0.1008
	Std	0.0239	0.0142	0.0224	0.0643	0.0207	0.0390	0.0225	0.0232	0.0128
ARI	Mean	0.3979	0.4051	0.1747	0.2183	0.0796	0.2131	0.3790	0.3794	0.0918
	Std	0.0252	0.0181	0.0189	0.0566	0.0235	0.0194	0.0319	0.0313	0.0138

Table 15.1. LED Display Domain, tree_depth = 3

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l^{0.5} \log(p_l)$	$b = -p_l \log(p_l)$
Accuracy	Mean	0.5616	0.5324	0.5487	0.5478	0.3038	0.3110	0.5148	0.5337	0.5395
	Std	0.0399	0.0357	0.0483	0.0468	0.0425	0.0357	0.0467	0.0342	0.0525
Precision	Mean	0.4861	0.4543	0.4718	0.4717	0.1990	0.2066	0.4324	0.4528	0.4632
	Std	0.0489	0.0518	0.0640	0.0611	0.0438	0.0481	0.0603	0.0505	0.0619
Recall	Mean	0.5616	0.5324	0.5487	0.5478	0.3038	0.3110	0.5148	0.5337	0.5395
	Std	0.0399	0.0357	0.0483	0.0468	0.0425	0.0357	0.0467	0.0342	0.0525
F1 score	Mean	0.5082	0.4749	0.4936	0.4928	0.2120	0.2134	0.4514	0.4740	0.4839
	Std	0.0465	0.0453	0.0578	0.0556	0.0410	0.0336	0.0549	0.0436	0.0597
ARI	Mean	0.3782	0.3494	0.3605	0.3609	0.1854	0.2054	0.3461	0.3538	0.3602
	Std	0.0403	0.0353	0.0480	0.0471	0.0454	0.0431	0.0460	0.0370	0.0458

Table 15.2. LED Display Domain, tree_depth = 4

Metric	Statistic	Entropy	$b = 1$	$b = p_l^{0.5}$	$b = \sqrt{p_l(1-p_l)}$	$b = p_l$	$b = p_l^2$	$b = -\log(p_l)$	$b = -p_l^{0.5} \log(p_l)$	$b = -p_l \log(p_l)$
Accuracy	Mean	0.6999	0.6735	0.6717	0.6711	0.3931	0.4246	0.6677	0.6732	0.6691
	Std	0.0290	0.0404	0.0438	0.0452	0.0324	0.0476	0.0533	0.0432	0.0458
Precision	Mean	0.7247	0.6937	0.6908	0.6920	0.3448	0.3895	0.6857	0.6907	0.6875
	Std	0.0361	0.0510	0.0546	0.0551	0.0498	0.0661	0.0687	0.0594	0.0599
Recall	Mean	0.6999	0.6735	0.6717	0.6711	0.3931	0.4246	0.6677	0.6732	0.6691
	Std	0.0290	0.0404	0.0438	0.0452	0.0324	0.0476	0.0533	0.0432	0.0458
F1 score	Mean	0.7022	0.6708	0.6682	0.6680	0.3207	0.3611	0.6626	0.6697	0.6653
	Std	0.0323	0.0473	0.0518	0.0519	0.0360	0.0543	0.0662	0.0524	0.0550
ARI	Mean	0.4485	0.4299	0.4304	0.4287	0.2644	0.2809	0.4269	0.4276	0.4281
	Std	0.0379	0.0430	0.0411	0.0454	0.0321	0.0307	0.0500	0.0462	0.0398